

# Current Work Summary – Sowmya Maddala

**Role:** *Machine Learning Engineer | Computer Vision & Time-Series Analysis*

**Duration:** May 2023 – Present

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## Key Contributions & Project Timeline

### 1. Computer Vision & Image Classification Projects

- **Fish Segmentation Model** (*May 2023 – Jun 2023*)
  - Developed instance segmentation models for fish using CNNs to enable automated detection in seafood processing.
- **Fresh/Spoilt Classification Using Feature Engineering** (*Jul 2023 – Oct 2023*)
  - Built models to classify individual fish as fresh or spoilt using image pixel feature engineering.
  - Applied this for multiple species.
- **Fish Species Classification & Multi-Species Segmentation** (*Nov 2023*)
  - Developed a 20-fish-species classification model and a common segmentation model for fish, prawns, and crabs.
  - Simultaneously led **data collection** and generated analytical reports using Python (Matplotlib, Seaborn).
- **Dataset Enhancement** (*Dec 2023*)
  - Classified fish into single/group, different backgrounds, and lighting conditions to create robust datasets.
- **Image Quality Classification** (*Jan 2024 – Feb 2024*)
  - Developed models to classify images as clear or blurred using CNNs and feature extraction.
- **Fresh/Spoilt Classification Using CNNs (Scratch & Transfer Learning)** (*Mar 2024 – Jun 2024*)
  - Implemented multiple CNN architectures on complete and segmented fish images.

- Used transfer learning for improved accuracy across datasets.
  - **Cuts & Damages Classification & Eye-Based Freshness** (*Jul 2024*)
    - Applied YOLO segmentation, feature extraction, and regression-based CNN classification for detecting fish cuts/damages.
  - **Model Optimization & Fine-Tuning** (*Aug 2024 – Sep 2024*)
    - Hyperparameter tuning, retraining with augmented datasets, experimenting with dimensionality reduction, custom metrics, and loss functions.
  - **3-Step Freshness Classification for Sardine Fish** (*Oct 2024 – Nov 2024*)
    - Introduced a three-step approach: 1) Cuts & damages, 2) Size, 3) CNN model classification.
  - **Advanced Segmentation & Automated Systems** (*Dec 2024*)
    - Built YOLO-based instance segmentation for detecting hair strands, worms, feathers, microplastic items.
    - Developed automated pipeline to detect and classify wheat, rice, and corn as good/spoilt.
    - Classified fish into fresh/spoilt based on sensor values for multiple species.
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## 2. Time-Series & Sensor Data Projects

- **Data Cleaning & EDA** (*Jan 2025 – Feb 2025*)
  - Sorted, cleaned, and analyzed multivariate sensor datasets for fish and fruit, generating detailed analytical reports.
- **Early Fish Degradation Alert System** (*Mar 2025 – Jul 2025*)
  - Preprocessed time-series data, handled missing timestamps offline and online.
  - Developed algorithms for **change point detection** using statistical methods and automatic thresholding.
  - Finalized change points using multiple sensor approvals.
  - Introduced new features:
    - Spoilage index labeling via mathematical modelling.

- Supervised classification of fresh/medium/spoilt fish.
  - Unsupervised clustering and outlier detection.
  - Time-to-spoil regression models.
  - Forecasting-based prediction using dynamic LSTM training.
  - **Fruit Dataset Exploration** (*Jul 2025*)
    - Applied similar approaches to fruit sensor datasets; identified limited usable sensors.
  - **Current Work** (*Aug 2025 – Present*)
    - Classifying fish as fresh/spoilt using sensor values.
    - Classifying freshness using wavelength and temperature data for multiple species.
    - Conducting EDA to derive insights for model improvement.
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### 3. Tools & Technologies

- **Computer Vision:** YOLOv8, CNNs, Instance Segmentation, Feature Extraction
  - **Time-Series Analysis:** Statistical Change Point Detection, Forecasting, Dynamic LSTM, Multivariate Sensor Fusion
  - **Machine Learning:** Supervised, Semi-Supervised, Unsupervised, Clustering, Regression, classification, transfer learning, fine-tuning
  - **Programming & Data Handling:** Python, NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, tensorflow, opencv
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### Summary

I have been applying **end-to-end machine learning and deep learning techniques** to real-world challenges in **seafood quality assessment and agriculture inspection**. My work spans **computer vision, sensor data analysis, time-series modelling, and real-time alert systems**, with a strong focus on **automated pipelines, model optimization, and actionable insights**. Currently, I am focused on **sensor-based fish freshness classification and wavelength-based multi-species analysis**.